

Ninad Amane

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PROFILE

AIML Engineering student with experience in Data Structures, Algorithms, Machine and Deep Learning. Skilled in building AI-based systems using Python, TensorFlow, PyTorch, Docker, and LLMs. Worked on NLP, federated learning, and autonomous coding projects, with two IEEE publications in NLP and medical imaging. Interested in problem-solving, competitive programming, and software engineering.

EDUCATION

BMS Institute of Technology & Management, B.E. in AI&ML

Bengaluru

- CGPA: **9.23**

Podar International School, Higher Secondary

Junagadh

- Class X: **91.6%** | Class XII: **89%**

WORK EXPERIENCE

AI&ML Intern, TestAlng.com

03/2026 – 05/2026

Bengaluru

- Performed error analysis on TRAI SMS classification models to identify confusion between promotional and service messages using precision, recall, F1-score, and confusion matrices.
- Researched multimodal sentiment analysis using text and image data, transformer models, deep learning, and Kaggle implementations.
- Explored OCR pipelines for the IIITK dataset, including image preprocessing, text extraction, OCR evaluation, and Python-based analysis.

SKILLS

- Programming Languages:** C/C++ | Python | HTML/CSS
- Core CS:** DSA | OOPs | DBMS | OS | CN | ToC
- NLP & ML:** Scikit-Learn | Neural Networks | ANNs, CNNs, RNNs, LSTMs, GRU | Feature Engineering | Text Embeddings | Model Deployment | Sentence Transformers | LLM Integration | Prompt Engineering
- Frameworks & APIs:** TensorFlow | Keras | NumPy | Pandas | Matplotlib | Seaborn | PyTorch | Streamlit | FastAPI | LangChain | LangGraph
- Databases:** SQL | MongoDB
- Cloud/Deployment:** Github | Postman | Docker | GitHub Actions | Cloud Computing

PROJECTS

Ghost Coder | Multi-Agent GitHub Issue Resolver 

- Architected a fully autonomous software engineering platform using LangGraph and Groq Llama-3, orchestrating a **3-agent state-machine** (Researcher, Coder, Tester) to independently analyze, debug, and resolve GitHub issues.
- Engineered a dynamic repository traversal system via PyGithub for targeted codebase context and integrated a secure Docker SDK sandbox, ensuring **100%** of AI-generated patches pass isolated automated execution before deployment.
- Developed a real-time observability dashboard in Streamlit with human-in-the-loop controls, leveraging the GitHub API to fully automate branch creation, staging, and pull requests, cutting manual deployment overhead by **80%**.

Neonatal Sepsis Detection using Federated Learning System 

- Implemented a privacy-preserving federated learning framework for early neonatal sepsis prediction, enabling secure multi-hospital EHR training using Flower and FedAvg, ensuring **HIPAA- and GDPR-compliant** collaboration.
- Developed GRU-D and Transformer time-series models on **40 NICU clinical features**, improving AUROC from **0.829 to 0.894** and accuracy from **73.9% to 94.7%** at optimized clinical decision thresholds to enhance early detection reliability.
- Strengthened clinical decision support by increasing precision from **0.187 to 0.712** (~280% relative increase) and reducing the false positive rate from **0.2422 to 0.0147**, minimizing alarm fatigue and unnecessary interventions.

Hybrid NLP Log Classification System 

- Created an NLP pipeline using semantic embeddings to automate log classification and intelligent incident routing across services.
- Improved classification accuracy by **40%** through structured LLM-based inference to handle previously misclassified edge-case logs.
- Reduced manual triage effort by **30%** using rule-based filtering to shorten response cycles for the engineering team.

PUBLICATIONS

Enhancing Question Answering Systems with Semantic Networks and NLP Techniques 

03/2025

Improving Medical Imaging Diagnostics with DCNs for Early Detection & Treatment 

03/2025

CERTIFICATES

- [Foundations of Deep Learning: Concepts and Applications, NPTEL](#) 
- [Oracle Cloud Infrastructure 2025 Certified Generative AI Professional](#) 
- [Machine Learning for Engineering and science applications, NPTEL](#) 
- [SnowPro Associate: Platform Certification](#) 
- [AWS APAC Solutions Architecture virtual Experience, Forage](#) 

ACHIEVEMENTS AND LEADERSHIP

- Secured the highest SGPA in the AIML department in the 3rd semester.
- Student Placement Coordinator assisting 70+ students with placement preparation.
- Solved 300+ DSA problems across LeetCode, GFG & CodeStudio.
- Presented 2 IEEE research papers at COMSIF Conference, BMSIT Bengaluru.
- Open Source Contributor at GirlScript Summer of Code 2026.